

COMPUTER SCIENCE DEPARTMENT

Privacy-Preserving Urban Acoustic Pattern Recognition for Smart City Emergency Detection

CS4507 – Pattern Recognition (Section 1)

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Abstract

The rapid expansion of smart city infrastructure necessitates robust emergency detection systems. However, traditional visual surveillance raises significant privacy concerns. This paper proposes a privacy-preserving acoustic pattern recognition system capable of detecting urban emergencies, specifically sirens and gunshots, using audio signals. The method transforms one-dimensional environmental audio into two-dimensional Mel-Spectrograms and treats acoustic classification as a spatial pattern recognition task.

To address the complexities of urban noise, the methodology restructures the traditional multi-class problem into a binary framework (**Emergency vs. Normal**), augmented by dynamic in-graph spatial transformations including

Random Zoom and Random Translation. Evaluated on the UrbanSound8K dataset, the proposed Convolutional Neural Network architecture achieved a validation accuracy of **97.02%**, significantly outperforming historical baselines.

1. Introduction

A. Problem Statement

As urban environments grow denser, the ability to rapidly detect and respond to emergencies becomes increasingly critical. Although computer vision systems are widely used in smart city deployments, they inherently compromise citizen privacy due to persistent visual monitoring. Acoustic pattern recognition offers a non-intrusive alternative; however, environmental sound classification in real-world urban environments remains challenging because of high background noise, overlapping signals, and acoustic variability.

B. Research Question and Hypothesis

Research Question: Can a binary-focused Convolutional Neural Network, utilizing spatially augmented two-dimensional Mel-Spectrograms, achieve higher accuracy and robustness in urban emergency detection compared to traditional multi-class acoustic models?

Hypothesis: By isolating emergency classes and applying dynamic in-layer data augmentations to simulate time-shifting and distance variations, the proposed model can extract more resilient spatial features and surpass standard classification baselines.

2. Related Work

Environmental sound classification has evolved substantially with the rise of deep learning. Earlier work by Piczak (2015) pioneered the transition from one-dimensional audio analysis to two-dimensional spatial learning through spectrogram-based CNNs, reporting a baseline accuracy of 73.1% on UrbanSound8K. Later, Salamon and Bello

(2017) emphasized the importance of explicit augmentation methods such as pitch-shifting and time-stretching, raising benchmark performance to 79.0%.

3. Proposed Method

A. Data-Level Novelty: Dynamic In-Graph Augmentation

Instead of pre-generating augmented audio samples offline, the proposed system applies dynamic spatial augmentation directly within the neural network graph using Keras layers.

- **Random Translation:** Shifts the Mel-Spectrogram matrix temporally and frequency-wise, simulating acoustic arrival delays and slight pitch displacement.
- **Random Zoom:** Scales the spectral image, simulating changes in source distance and Doppler-like variation.

B. Methodological Novelty: Strategic Class Binarization

Traditional models classify all ten UrbanSound8K classes simultaneously, which commonly leads to confusion among acoustically similar background sounds. This project reformulates the dataset into a binary paradigm tailored for smart city emergency applications:

- **Emergency (Class 1):** Sirens (Class 8) and Gunshots (Class 6)
- **Normal (Class 0):** All remaining ambient classes, including dog barking, street music, and jackhammers

C. Feature Extraction Pipeline

Raw audio signals are converted into Mel-Spectrogram matrices. The Mel scale is a perceptual scale of pitch that better reflects human hearing, and it is mathematically expressed as:

$$m = 2595 \log_{10}(1 + f / 700)$$

This conversion maps each audio sample into a $128 \times 128 \times 1$ grayscale spatial matrix, enabling the CNN to perform structural pattern recognition on acoustic data.

D. CNN Architecture

The proposed model is a deep sequential CNN consisting of three main convolutional blocks. Each block uses 3×3 kernels, followed by Batch Normalization, MaxPooling2D, and Dropout with a rate of 0.25 to reduce overfitting. The extracted spatial features are flattened and passed through a Dense layer with 128 units, ending with a Sigmoid activation function for binary prediction.

4. Experiments and Evaluation

A. Dataset Setup

The UrbanSound8K dataset, containing 8,732 labeled sound excerpts of up to four seconds each, was used in this study. Audio files were processed using the Librosa library and resampled at 22,050 Hz. The data was split into 80% training and 20% validation sets.

B. Implementation Details

- **Optimizer:** Adam optimizer for adaptive learning rate computation
- **Loss Function:** Binary Crossentropy
- **Hyperparameters:** Batch size of 32, training for 20 epochs, and execution on an NVIDIA T4 GPU using Google Colab
- **Callbacks:** ModelCheckpoint to save the optimal weights based on validation accuracy

5. Results

The proposed augmented architecture demonstrated excellent learning stability and generalization performance.

- **Training Performance:** The model converged efficiently and reduced binary cross-entropy loss to 0.1031.
- **Validation Accuracy:** The model achieved a peak validation accuracy of 97.02% on unseen validation samples.

Model / Methodology	Classification Type	Accuracy
Piczak (Baseline)	Multi-class (10)	73.10%
Salamon & Bello	Multi-class (10)	79.00%
Our Proposed Method	Binary (Emergency Focused)	97.02%

6. Discussion and Limitations

The substantial performance gain is primarily attributed to the methodological decision to binarize the target variable. By removing the burden of distinguishing among multiple non-emergency sounds, the CNN can concentrate its filters on identifying the characteristic spectral signatures of sirens and the impulsive signatures of gunshots. Furthermore, in-graph augmentation improves robustness against simulated distance variation and temporal misalignment.

Despite the high accuracy, the system remains susceptible to roughly 3% false negatives. These cases typically arise in masked-noise scenarios, where an emergency sound is heavily overshadowed by louder ambient noise occurring near the microphone at the same time.

7. Conclusion

This project successfully developed a privacy-first urban acoustic monitoring system for emergency detection. By converting linear audio into two-dimensional spatial features and using a deep, augmented Convolutional Neural Network, the system achieved a highly reliable emergency detection rate of 97.02%. The integration of

robust data augmentation and binary problem formulation proved highly effective for smart city applications while maintaining ethical surveillance constraints.

8. Contribution Statement

- **Faris Alsulami:** Led the CNN architectural design, implemented the in-graph data augmentation strategies, and developed the final interactive Streamlit deployment dashboard.
- **Hussain Mash:** Managed dataset acquisition, metadata restructuring for binary classification, and conducted the model training experiments.
- **Abdulrahim Alharbi:** Managed the feature engineering pipeline using Librosa, performed performance evaluation, and contributed to the literature review and final report formatting.

9. References

1. J. Salamon, C. Jacoby and J. P. Bello, “A Dataset and Taxonomy for Urban Sound Research,” 22nd ACM International Conference on Multimedia, Orlando, FL, USA, 2014.
2. K. J. Piczak, “Environmental sound classification with convolutional neural networks,” 2015 IEEE 25th International Workshop on Machine Learning for Signal Processing (MLSP), Boston, MA, 2015, pp. 1–6.
3. J. Salamon and J. P. Bello, “Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification,” IEEE Signal Processing Letters, vol. 24, no. 3, pp. 279–283, March 2017.